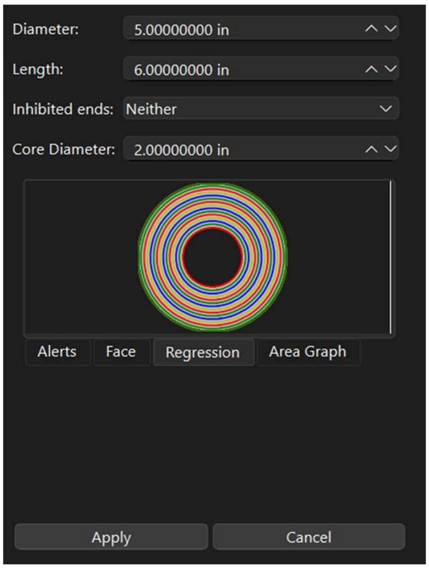
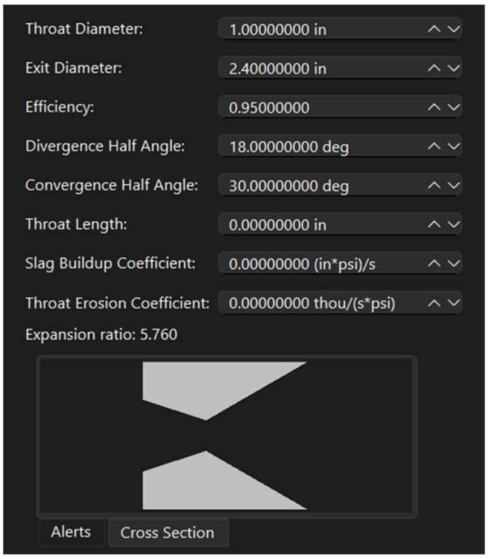
OpenMotor

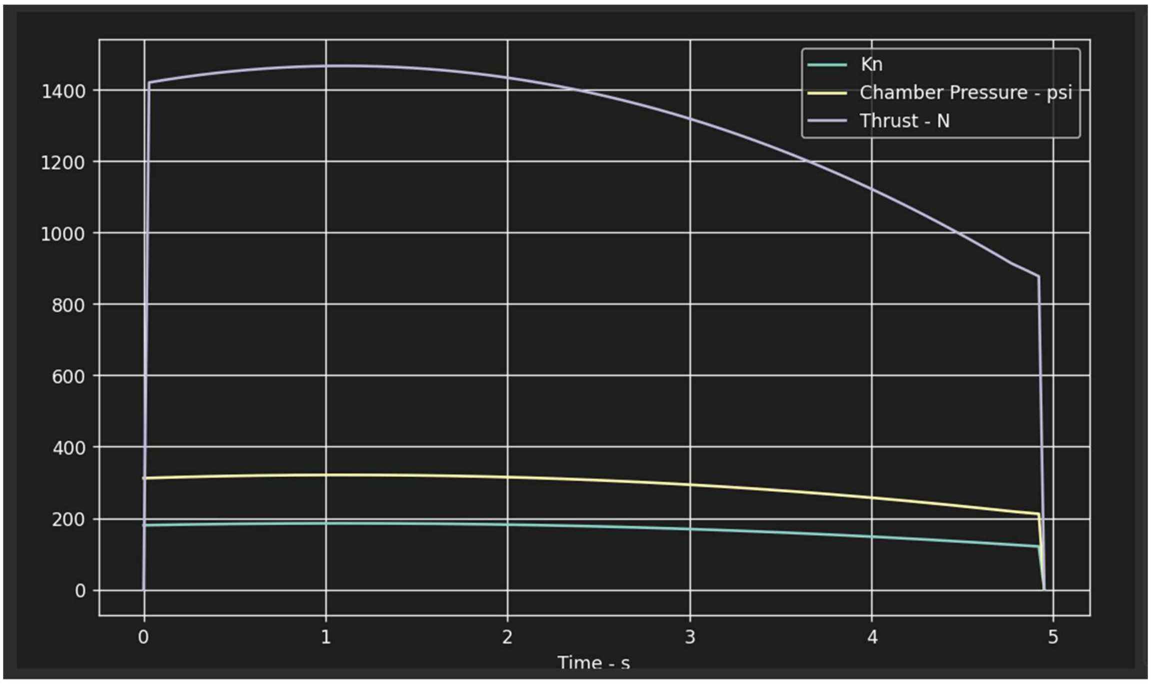
Our first rocket features a Solid Rocket Motor (SRM) using Potassium Nitrate/Sorbitol (KNSB) propellant, with a 65:35 mass ratio. It employs two Bates-pattern fuel grains, each with a 5-inch outer diameter, 2-inch core diameter, and 6-inch height. The nozzle, designed to optimize exhaust velocity, has a 30° convergence angle, 18° divergence angle, 1-inch throat diameter, 2.4-inch exit diameter, and zero throat length for simplified manufacturing. The nozzle is constructed from heat-resistant graphite (throat) and linen phenolic, housed in an aluminum nozzle carrier.

Calculations and observations:

Using these values we get the following results when simulating in OpenMotor:





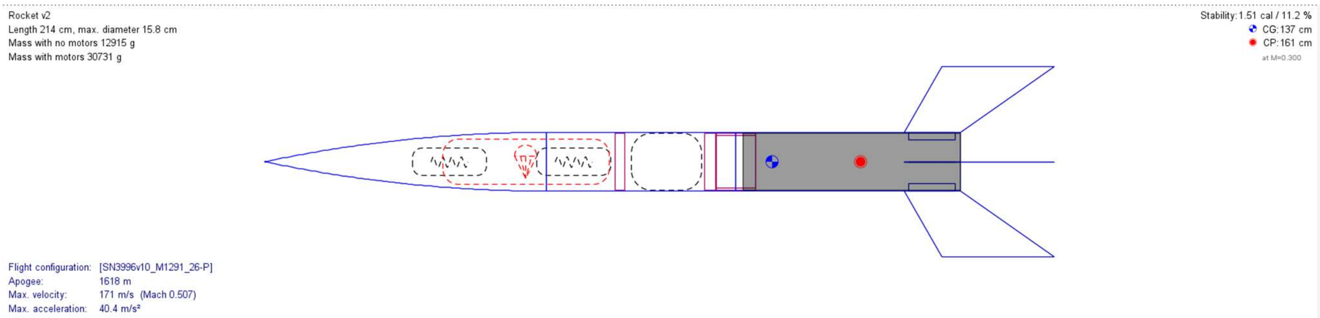


Motor Statistics

Motor Designation	M1291 (26%)	Average Pressure	289.7 psi	Propellant Mass	5.68 kg
Impulse	6432.11 Ns	Peak Pressure	320.98 psi	Propellant length	12.00 in
Delivered ISP	115.56 s	Initial Kn	180.00	Port/Throat Ratio	4.00
Burn Time	4.95 s	Peak Kn	185.33	Peak Mass Flux	0.74 $lb/in^2.s$
Volume Loading	84.00%	Ideal Thrust Coefficient	1.40	Delivered Thrust Coefficient	1.28

OpenRocket

Based on the CAD file (explained in a separate file), the Rocket was designed in OpenRocket simulation software, ensuring that the design in openrocket is as close as possible to the design in the CAD file. The simulation results are as follows:



Key statistics:

- Velocity off rod: 8.34 m/s
- Apogee: 1618 m
- Optimum delay: 14.8 s
- Max velocity: 171 m/s
- Max acceleration: 40.4 m/s^2
- Time to apogee: 19.8 s

Some more talking points

1. The Rocket has 6 inch diameter, which is larger than usual for a rocket that reaches 1km apogee. We have chosen this diameter keeping in mind the motor’s reusability in future projects where apogee may reach more than 5 kilometres.
2. The Nozzle throat is 1 inch in diameter, which results in not so efficient conversion of thermal energy to kinetic energy. But decreasing the throat diameter constricts the flow, causing average and peak pressure to rise significantly high. Keeping in mind that this is the first rocket, we have chosen this specific nozzle design and have optimized it so that the rocket reaches an apogee of 1 km.
3. The overall weight of the rocket is around 30 kg, which might seem heavier than usual 1km-apogee model rockets. But again, as we emphasise on reusability of Rocket parts for future bigger projects, we have proceeded with this design and optimized the weight for 1 km apogee. We are still looking for options to reduce more weight in this design for better efficiency.